WHICH ARRHENIUS INTERVAL TO TRUST? TWO-STAGE E_a RECOVERY, EQUAL INFORMATION

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ABSTRACT

Activation energies from first-order decay measured at several temperatures are reported with a 95% interval, but its coverage depends on choices that are rarely tested together: how per-temperature rate-constant errors become regression weights, and how the interval is built. We calibrate the two-stage Arrhenius estimator (nonlinear least squares for k at each temperature, then weighted regression of $\ln k$ on centred $1/T$) by Monte Carlo against Cramér–Rao and analytic references: five temperatures, k spanning $10\times$, time grids scaled to $3/k$ so that every temperature is equally informative, five noise levels, five sample sizes, 2000 replicates per cell. In this design equal weights are the true weights and give $\mathrm{sd}(\hat{E}_a)$ of 0.97–1.03 times the bound for noise up to 10% of c_0; estimating the weights from each fit raises it by 4–14% ($\sigma \le 0.1c_0$, $n \le 12$) and lowers propagated-z coverage to 0.84–0.88 at $n = 5$. A propagated t interval with equal weights (0.95–0.97 at $n = 5$) or a residual-scaled t_3 interval (0.94–0.96) holds near nominal given convergence; at 20% noise and $n = 5$, where 4.2% of replicates have a boundary fit, their unconditional coverage is 0.925 and 0.899. Under one $4\times$ noise ramp the ranking of weights reverses (1.27 times the bound with equal weights, 1.02–1.08 with estimated weights) while residual-scaled t_3 coverage stays 0.94–0.96.

1 INTRODUCTION

A first-order reaction $c(t) = c_0 e^{-kt}$ with Arrhenius rate $k = Ae^{-E_a/(RT)}$ is often characterised by fitting k at several temperatures and regressing $\ln k$ on $1/T$. The slope gives E_a, the intercept $\ln A$, and the regression standard error gives the reported interval. Whether that interval contains the truth 95% of the time depends on how the per-temperature standard errors become regression weights, on whether the interval uses propagated or residual scatter, and on whether n is large enough for each stage-one standard error.

We simulate this estimator with the truth known and the Cramér–Rao bound (CRB) as reference, and report bias, spread and interval coverage as the noise level and the number of time points n vary. In an equal-information design (time grids scaled to $3/k$ at every temperature), equal weights reach the CRB within about 3% for noise up to 10% of c_0 and a propagated t or residual-scaled t_3 interval holds near 95% down to $n = 5$; per-temperature $1/\widehat{\mathrm{se}}^2$ weights with a z interval fall to 0.84–0.88 (Figure 3), and the ranking of weight choices reverses under a noise ramp. The contributions are:

1. A map of $\mathrm{sd}(\hat{E}_a)/\mathrm{CRB}$ and of coverage over the 5×5 noise-by-n grid (Figures 1 and 3, Table 1), including the cost, up to 13.8% in spread, of estimating the weights per temperature.
2. An explanation: the dimensionless time kt makes every temperature equally informative, so equal weights are the true weights and the estimated weights add noise.
3. The cost of the $3/k$ observation horizon against the Fisher optimum, 12–18% in $\mathrm{sd}(\ln \hat{k})$ (Figure 2).

2 SETUP AND REFERENCES

Data. Five temperatures $T_i = 300.0$, 308.9, 318.3, 328.3, 338.9 K are equally spaced in $1/T$, with $E_a = 50\,\mathrm{kJ\,mol^{-1}}$, $R = 8.314\,\mathrm{J\,mol^{-1}K^{-1}}$ and $k_i = 0.010$, 0.018, 0.032, 0.056, 0.100 (a

$10\times$ span; $\ln A = 15.44$). At temperature i, n times are equally spaced on $[0, 3/k_i]$ including $t = 0$, and $y_{ij} = c_0 e^{-k_i t_{ij}} + \varepsilon_{ij}$ with $c_0 = 1$, $\varepsilon_{ij} \sim \mathcal{N}(0, \sigma^2)$. We use $\sigma \in \{0.01, 0.02, 0.05, 0.1, 0.2\}$ and $n \in \{5, 8, 12, 20, 40\}$, with R=2000 replicates per cell; c_0 is known except in the four cells of the c_0-fitted check in Section 3. The grid in $u = kt$ is the same at every T, so the Fisher information about $\ln k_i$ is equal across temperatures.

Stage 1. At each T_i we minimise the residual sum of squares over k (grid search over $\ln k$, then local refinement) and take $\widehat{\mathrm{se}}_i$ from the Gauss–Newton information with $\hat\sigma^2$ on $n - 1$ degrees of freedom. A fit whose minimum sits on the search boundary is a failure and is counted (fraction *Fail*); statistics are reported *given convergence* (cond) and, where stated, with failures counted as misses (uncond).

Stage 2. With $x_i = 1/T_i$ and weights w_i we fit $\widehat{\ln k_i} = a + b\,x_i$ by weighted least squares and set $\hat E_a = -Rb$, $\widehat{\ln A} = a - b\bar x$ ($\bar x$ the mean of x_i; the line is fitted on $x_i - \bar x$). Weights are *equal* ($w_i = 1$), *est* ($1/\widehat{\mathrm{se}}_i^2$, the usual choice), *pool* ($1/\widehat{\mathrm{se}}_i^2$ with one $\hat\sigma$ pooled over the five fits) or *true*. The 95% intervals are *propagated*, $\hat E_a \pm c\,R\sqrt{\sum_i g_i^2 \widehat{\mathrm{se}}_i^2}$ with $g_i = \partial b/\partial \widehat{\ln k_i}$ (valid for any weights) and $c = 1.96$ (z) or the t_ν quantile with Satterthwaite ν ($5(n-1)$ for pool), or *residual-scaled*, $\hat E_a \pm t_{3,0.975} R\,s/\sqrt{\sum_i w_i (x_i - \bar x_w)^2}$ with $s^2 = \sum_i w_i r_i^2/3$, r_i the residuals of $\widehat{\ln k_i}$ about the line and $\bar x_w$ the weighted mean of x_i.

References. (i) CRB: for c_0 known the five fits are independent with information $\sum_j (u_j e^{-u_j})^2/\sigma^2$ about $\ln k$, so $\mathrm{CRB}(\hat E_a) = R\,\mathrm{CRB}(\ln \hat k)/\sqrt{\sum_i (x_i - \bar x)^2}$; in this design the equal weights *are* the true weights. (ii) A residual-scaled z interval has coverage $P(|t_3| < 1.96) = 0.855$ when the stage-one errors are exactly normal. (iii) For the equal-weight line, $\mathrm{corr}(\hat E_a, \widehat{\ln A}) = \bar x/\sqrt{\bar x^2 + s_x^2} = +0.9991$ with $s_x^2 = \frac{1}{5}\sum_i (x_i - \bar x)^2$, which follows from extrapolating to $1/T = 0$; we use it as a code check. MC SEs of an sd use the kurtosis-corrected formula; coverage SEs are binomial. Random numbers come from `numpy.random.SeedSequence(20261015)`; `kinetics_mc.py` runs in 20.9 s on a CPU and writes every number below to `results.json`.

Related work. Reparametrising the Arrhenius equation around a reference temperature to decorrelate $\ln A$ and E_a is treated by Schwaab & Pinto (2007) and Schwaab et al. (2008) (records verified via Crossref; full texts not read). Estimating weights in weighted least squares is standard; the quantity reported here is the size of the resulting loss in spread and coverage in a kinetic design.

3 Results

Noise σ is in units of $c_0 = 1$. Code checks: with $\sigma = 10^{-6}$ the error in $\hat E_a$ is 2×10^{-11} J mol^{-1}, and at $\sigma = 0.01$, $n = 40$ the equal-weight sd/CRB is 0.980 ± 0.011 (4000 replicates); these test the code and are not findings.

Spread (Figure 1, Table 1). With equal weights $\mathrm{sd}(\hat E_a)/\mathrm{CRB}$ lies between 0.973 and 1.032 (min–max over cells, MC SE 0.016, so within two SE of 1) for every cell with $\sigma \le 0.1$. At $\sigma = 0.2$ it is 1.113 ± 0.020 at $n = 5$, 1.063 ± 0.026 at $n = 12$ and 0.992 ± 0.016 at $n = 40$; at $n = 5$ the fraction of replicates with a boundary fit is 0.042, and the values are conditional on convergence. Estimating the weights per temperature increases the spread for $\sigma \le 0.1$: by 9.9–13.8% at $n = 5$ (largest at $\sigma = 0.02$; 1.151 against 1.032 ± 0.016, 11.6%, at $\sigma = 0.01$) and by 4–9% at $n = 8$ and 12. Pooling $\hat\sigma$ over temperatures removes the increase for $\sigma \le 0.1$. At $\sigma = 0.2$ the pooled spread is 0.956–1.02, below the equal-weight spread at small n (0.963 against 1.113 at $n = 5$). The weights are equal in truth, so for $\sigma \le 0.1$ estimating them only adds variance, and that variance falls as n rises.

Coverage (Figure 3). All coverages below are conditional on convergence unless tagged uncond. With equal weights the propagated-t interval covers 0.9545–0.966 at $n = 5$ across all σ and the residual-scaled t_3 interval covers 0.939–0.958 over all 25 cells (binomial SE ≤ 0.006). The residual-scaled z interval covers 0.840–0.874, as predicted by the analytic 0.855. With estimated weights the

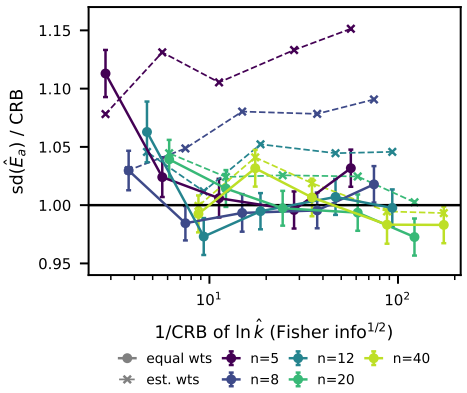


Figure 1: $\mathrm{sd}(\hat{E}_a)/\mathrm{CRB}$ against $1/\mathrm{CRB}(\ln \hat{k})$ (square root of the Fisher information of $\ln k$ at one temperature); noise σ/c_0 falls from 0.2 (left) to 0.01 (right) within each n. Solid: equal weights; dashed: estimated weights. R=2000; bars ± 1 MC SE on the equal-weight points.

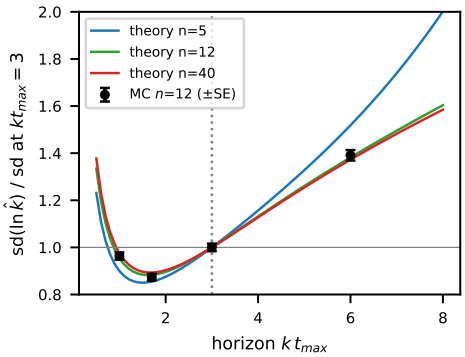


Figure 2: sd of $\ln \hat{k}$ at a single temperature against horizon $k\,t_{\max}$, relative to the $k\,t_{\max}$=3 design. Lines: Fisher theory, $n = 5, 12, 40$ (optimum 1.5–1.7, sd 0.85–0.89 of the $3/k$ design). Points: Monte Carlo, n=12, σ/c_0=0.05, 4000 replicates, ±SE within markers.

Table 1: Selected cells (R=2000; 9 of 25, spanning the grid). sd/CRB: sd of $\hat{E}_a$ over the Cramér–Rao bound, c_0 known, given convergence; $\pm$ is the MC SE (equal-weight column). eq.: equal weights; est.: estimated weights; pool: pooled σ. Coverage of E_a by nominal 95% intervals given convergence; z, t_ν: propagated; t_3: residual-scaled; unc.: failures counted as misses. Fail: fraction of replicates with a boundary fit.

σ	n	sd/CRB			coverage					Fail
		eq.	est.	pool	z eq.	z est.	t_ν eq.	t_3 eq. (unc.)	t_3 est.	
0.01	5	1.032±0.016	1.151	1.032	0.930	0.843	0.955	0.949 (0.949)	0.924	0.000
0.02	5	0.996±0.016	1.133	0.994	0.933	0.854	0.957	0.952 (0.952)	0.923	0.000
0.02	40	0.983±0.016	0.995	0.983	0.947	0.943	0.951	0.949 (0.949)	0.948	0.000
0.1	5	1.024±0.017	1.131	0.994	0.924	0.858	0.956	0.953 (0.953)	0.930	0.000
0.1	12	0.973±0.016	1.012	0.963	0.955	0.926	0.961	0.958 (0.958)	0.950	0.000
0.1	40	1.032±0.016	1.041	1.031	0.941	0.933	0.941	0.947 (0.947)	0.945	0.000
0.2	5	1.113±0.020	1.078	0.963	0.936	0.877	0.966	0.939 (0.899)	0.927	0.042
0.2	12	1.063±0.026	1.046	1.004	0.943	0.929	0.951	0.954 (0.954)	0.946	0.000
0.2	40	0.992±0.016	0.999	0.983	0.949	0.948	0.951	0.946 (0.946)	0.943	0.000

propagated-z interval covers only 0.843–0.877 at $n = 5$ (0.841–0.858 uncond), against 0.92–0.94 for propagated z with equal weights; relative to 0.95, equal-weight propagated z loses about two points and estimating the weights from five fits about eight more. Failures matter in one cell: at $(\sigma, n) = (0.2, 5)$ equal-weight propagated-t coverage is 0.966 cond but 0.925 uncond, and residual-t_3 0.939 cond but 0.899 uncond (0.927 and 0.888 with estimated weights); in every other cell with c_0 known cond and uncond differ by at most 0.002. For $\ln A$ the equal-weight sd/CRB spans 0.973–1.113 and propagated-t coverage 0.941–0.967 over the grid.

Bias. $\hat{E}_a$ bias exceeds two MC SE in 3 of 25 cells (about 1.25 expected by chance; largest -470 ± 170 J mol^{-1}, -0.9% of E_a). The bias of $\ln k$ at the reference temperature is larger relative to its SE: at $(0.2, 5)$ it is $+0.030$ at stage one, $+0.030 \pm 0.004$ with equal weights and -0.040 ± 0.004 with estimated weights.

Correlation and horizon. $\mathrm{corr}(\hat{E}_a, \widehat{\ln A})$ is $+0.9990$–$+0.9991$ in every cell (computed with estimated weights), matching the equal-weight identity $+0.9991$ (an algebra and code check). For a single temperature with c_0 known (Figure 2) the Fisher optimum horizon is $k\,t_{\max} = 1.50, 1.62, 1.67$ for $n = 5, 12, 40$, and the $3/k$ design used above costs $+17.6\%$, $+13.3\%$, $+12.0\%$ in $\mathrm{sd}(\ln \hat{k})$. A Monte Carlo at $n = 12$, $\sigma = 0.05$ (4000 replicates) agrees with the CRB at $k\,t_{\max} = 1, 1.7, 3, 6$ to within about 2%.

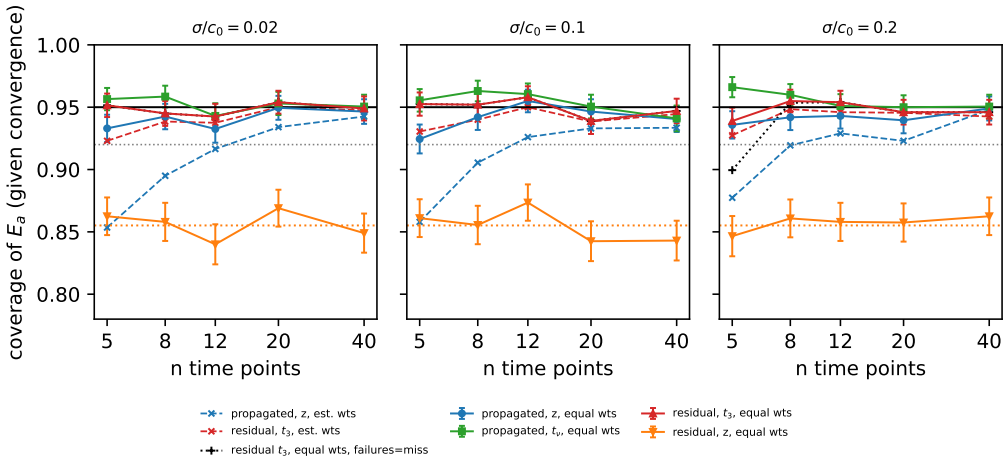


Figure 3: Coverage of E_a by nominal 95% intervals against n (given convergence), for 3 of the 5 noise levels. Four constructions with equal weights (solid): propagated z, propagated t_ν, residual-scaled z, residual-scaled t_3. Dashed: propagated z and residual t_3 with estimated weights. Dotted black: residual t_3, equal weights, failed replicates counted as misses; it coincides with the solid curve except at $(\sigma/c_0, n) = (0.2, 5)$. Bars are 95% binomial intervals; black line, nominal 0.95; orange dotted, analytic $P(|t_3| < 1.96) = 0.855$; gray dotted, 0.92.

Other checks. *Heteroscedastic noise.* One $4\times$ ramp of σ_i across temperatures ($\sigma = 0.02$, $n = 8$ and 20) gives 1.27–$1.28\times$ the CRB with equal weights, 0.98–1.00 with true weights, 1.02–1.08 with estimated weights and 1.26–1.27 with pooled $\hat{\sigma}$; residual-t_3 coverage is 0.941–0.958 with equal and 0.938–0.942 with estimated weights. *c_0 fitted.* In four cells ($\sigma \in \{0.05, 0.2\}$, $n \in \{8, 20\}$, 2000 replicates) equal-weight sd/CRB is 1.014–1.120 and residual-t_3 coverage 0.941–0.954; ten further c_0-fitted cells with 500 replicates are not used for claims. *Seeds.* Over three master seeds the largest change in residual-t_3 coverage (estimated weights) in any cell is 0.019.

4 Discussion, Limitations and Conclusion

In the equal-information design with a correct model, homoscedastic Gaussian noise and known c_0, use equal weights (or pooled $\hat{\sigma}$) with a propagated t or residual-scaled t_3 interval. Equal weights are the true weights because $u = kt$ is identical at every temperature. When the noise level differs across temperatures the choice flips: under the $4\times$ ramp equal weights give $1.27\times$ the CRB and estimated weights 1.02–1.08, so the recommendation needs homoscedastic noise.

Limitations. (1) The data are generated by the fitted model with additive homoscedastic Gaussian noise and time grids scaled to $3/k$ at each temperature; multiplicative or heavy-tailed noise, model misspecification, a fixed absolute time window across temperatures, a one-stage global fit, and other numbers of temperatures or spans were not run. (2) One five-temperature design is used, and the heteroscedastic and c_0-fitted results are one ramp and four cells, so they are indications, not maps. (3) The cause of the $\ln k$ bias at the reference temperature is untested. (4) The two references in Section 2 were not read in full, and no independent integrity audit was run.

Conclusion. In a correctly specified, equal-information Arrhenius design, an equal-weight two-stage fit reaches the Cramér–Rao spread for E_a within about 3% for $\sigma \leq 0.1c_0$, and a propagated t or residual-scaled t_3 interval stays near 95% given convergence down to $n = 5$; estimating the weights per temperature degrades both, unless noise differs across temperatures.

References

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